WEEK 1 · LESSON 4 OF 24 · HANDOUT

What AI Does Well — and Where It Fails

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=cfqtFvWOfg0 — open the lesson online to watch inside the course.

🎓 **Go deeper:** CS50 Seminar: How to Predict the Future with AI, One Word at a Time — https://www.youtube.com/watch?v=NGEEN0BzH-s

Your goal in this lesson: List the main strengths and failure modes of AI tools, recognise hallucinations and bias, and decide when a task is safe to hand to AI.

The jagged edge

AI is surprisingly good at some hard tasks and surprisingly bad at some easy ones. It can write a decent poem in seconds but may miscount the letters in a word or invent a book that does not exist. Researchers call this a **jagged frontier**: there is no smooth line between "easy for AI" and "hard for AI". You learn where the edge is by trying, checking and noticing.

What AI does well

- **Drafting** — first versions of emails, reports, letters, lesson plans.
- **Summarising** — turning 20 pages into one page of key points.
- **Explaining** — breaking difficult ideas into simple steps, at the level you ask for.
- **Transforming** — changing tone, translating, turning notes into a table.
- **Brainstorming** — giving you many options quickly to choose from.
- **Pattern spotting** — finding trends in data you provide.

Where AI fails

1. **Hallucinations** — invented facts, fake quotes, made-up references and links that look real.
2. **Out-of-date knowledge** — the model may not know recent events, new laws or new prices.
3. **Local blind spots** — less training data about Rwanda, Kinyarwanda and local institutions means more mistakes on local topics.
4. **Bias** — patterns learned from unbalanced data can treat some groups unfairly.
5. **Maths and exact counting** — many tools still make arithmetic slips unless they use a calculator tool.
6. **Over-confidence** — the tone sounds certain even when the content is a guess.
7. **No responsibility** — the AI cannot be held accountable. You can.

A quick risk test before you use AI

Ask yourself two questions:

	Low cost if wrong	High cost if wrong
Easy for me to check	✅ Use AI freely (brainstorming, drafts)	⚠️ Use AI, then check carefully (reports, figures)
Hard for me to check	⚠️ Use AI as a starting point only	⛔ Do not rely on AI (medical, legal, financial decisions)

Spotting a hallucination

Warning signs:

- Very specific numbers, dates or names without a source.
- References or links you cannot find when you search for them.
- Quotes from real people that you cannot confirm.
- An answer that changes when you ask the same question again.

Simple habit: for anything that matters, ask the AI *"What are your sources?"*, then check those sources yourself.

Remember this: Let AI do the drafting, summarising and explaining — and keep human judgement for facts, fairness and final decisions.

Name: _____ Date: _____

Activity

Task 1

Ask an AI assistant for *"three academic books about digital transformation in Rwanda, with authors and years."*

Task 2

Search for each book online or in a library catalogue.

Task 3

Record which ones are real, which are partly wrong, and which seem invented. What does this teach you about using AI for references?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does the 'jagged frontier' of AI mean?
   A. AI only works in some countries
   B. AI is equally good at everything
   C. AI can be strong at some hard tasks and weak at some easy ones
   D. AI tools have sharp graphics

2. Which task is AI generally GOOD at?

A. Taking legal responsibility for a decision
B. Guaranteeing every reference is real
C. Summarising a long document into key points
D. Knowing tomorrow's exchange rate

3. Which is a warning sign of a hallucination?

A. The answer uses simple words
B. The answer is short
C. The AI asks a follow-up question
D. A reference you cannot find anywhere when you search

4. Why might AI make more mistakes on local Rwandan topics?

A. There is usually less training data about very local topics
B. AI cannot read place names
C. Rwanda bans AI tools
D. Local topics are too easy

5. A task is HIGH cost if wrong and HARD for you to check (for example, a medical decision). What should you do?

A. Rely fully on the AI
B. Ask the AI twice and choose the longer answer
C. Use AI because it sounds confident
D. Do not rely on AI; use qualified experts

6. Who is responsible for work you submit that used AI?

A. The AI tool itself
B. You
C. The AI company
D. Nobody

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